GEK-36538

INSTRUCTIONS

POWER SYSTEM STABILIZER

3S7932LA202G5 & G6

NOTE: Includes speed transducer for toothed wheel and magnetic pickup.

GENERAL ELECTRIC

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INTRODUCTION

The Power System Stabilizer (PSS) is designed to introduce a supplementary control signal into continuously acting voltage regulator equipment. The equipment will supplement the conventional voltage regulator control with a control function derived so that, under deviations of generator speed or load, excitation becomes controlled as a composite function of voltage and shaft speed. Under certain conditions, the composite function has been shown to be an aid to power system stability.

The PSS receives a signal from a toothed wheel with magnetic pick-up located at the turbine front standard. This signal, which is frequency modulated, is passed through a speed transducer which then has an output frequency equal to the modulation frequency and an amplitude proportional to the deviation in frequency. This signal is then passed through two lead-lag stages in order to obtain the proper phase shift characteristic, a washout stage to eliminate the effect of component drift and slow system frequency variations, a gain stage to obtain maximum control action and an output stage which interfaces the PSS with continuously acting voltage regulating equipment.

The Power System Stabilizer includes the functions of input transducer (not including the toothed-wheel magnetic pickup), signal conditioning, limiting and output circuitry. Figure 4 is a typical functional elementary diagram while Figure 5 is composed of the printed circuit board elementary diagrams.

The Power System Stabilizer will apply the following transfer function to modify the output of the speed transducer for insertion into the voltage regulator:

$$G(s) = \frac{K\ (T_1S)\ (1 + T_2S)\ (1 + T_4S)}{(1 + T_1S)\ (1 + T_3S)\ (1 + T_5S)}$$

Adjustments are provided to vary the parameters of the PSS so as to permit compensation for variations in the characteristics of the regulator, excitation system, generator and power system to which it is applied. These adjustments can be made by calibrated dial settings which will be accurate to within 10% of the desired parameter value. The adjustments are capable of modifying the transfer function parameters through the following typical ranges:

K = 1 to 100

T_1 = 0.1 to 50 seconds

T_2 = 0.2 to 1.5 seconds

T_3 = 0.02 to 0.1 seconds

T_4 = 0.2 to 1.5 seconds

T_5 = 0.02 to 0.1 seconds

RECEIVING AND HANDLING

The Power System Stabilizer is normally supplied mounted in either the regulator or exciter cubicle. The instruction manual for the particular excitation system used should be consulted for receiving and handling instructions. If the PSS is shipped separately (as in the case of a retrofit system), then immediately upon receipt, the equipment should be carefully unpacked to avoid damage. It should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric sales office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If equipment is to be stored for more than three months, it should be kept heated to at least 5°C above ambient temperature to prevent condensation.

ABBREVIATIONS

The following abbreviations are used in this instruction manual:

ac	Alternating Current
Amp	Amperes
Db	Decibels
dc	Direct Current
FET	Field Effect Transistor
Hz	Hertz
IC	Integrated Circuit
Ma	Milliamp
Mv	Millivolt
PP	Peak-to-Peak
PSS	Power System Stabilizer
SCR	Silicon Controlled Rectifier
VAC	Alternating Current Voltage
VDC	Direct Current Voltage

DESCRIPTION (COMPONENTS)

The Power System Stabilizer consists of:

3S7932LA202G5	Panel	OR
3S7932LA202G6	Rack Assembly	

The Power System Stabilizer is normally mounted in either the regulator cubicle or exciter cubicle and contains the following:

Number	Name	Quantity
44C372607-G01	4800 Hz Speed Transducer	1
44C331857-G01	Signal Conditioning	1
44C331859-G01	Washout and Output	1
44C331863-G01	Relay	1
44C331867-G01	Ceiling Sensing	1
*44C331862-G01	Power Supply	1
*3S7932MA321G8	Terminal Voltage Sensing Panel	1
*3S7932MA321G9	± 36 Volt Power Supply Panel	1
*3S7932MA321G10	24 Volts Power Supply Panel	1

*Not furnished with 3S7932LA202G6

INSTALLATION

LOCATION AND CONNECTIONS

The Power System Stabilizer is normally furnished as an integral part of the excitation and regulation equipment. (Refer to the drawings furnished with the excitation system and the instruction manual for the excitation system for location and connection instructions.)

INITIAL CHECKS

1. Wire check all external connections to the Power System Stabilizer. Be sure that wire size and type are adequate for the expected signal levels. (Refer to the connection diagram furnished with the excitation equipment.)

INSTALLATION (continued)

INITIAL CHECKS (continued)

CAUTION

DO NOT HI-POT OR MEGGER ANY CIRCUIT, AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE PSS.

2. Insure that all printed circuit boards are fully inserted.

3. Insure that the shield on the cable from the magnetic pick-up is grounded only at the excitation equipment.

OPERATION

CAUTION

MISOPERATION OF THE POWER SYSTEM STABILIZER WHILE CONNECTED TO THE EXCITATION SYSTEM MAY RESULT IN DAMAGE TO THE GENERATOR. DEVIATIONS FROM THE FOLLOWING OPERATIONAL CHECKS AND PROCEDURES SHOULD BE CONSIDERED IN LIGHT OF THIS AND SHOULD BE DISCUSSED WITH THE MANUFACTURER PRIOR TO INITIATION.

INITIAL OPERATION

CAUTION

THE POWER SYSTEM STABILIZER MUST BE OUT OF SERVICE OR THE REGULATOR IN THE MANUAL MODE FOR ALL TESTS THROUGH PARAGRAPH 9 (EXCEPT PARAGRAPH 3 AS NOTED). FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

1. With the turbine generator at rated speed, check the output of the magnetic pickup with an oscilloscope. This can be done by connecting the oscilloscope leads across terminals L23 and L24 or connecting the leads to "IN TEST" on the Speed Transducer board (LOC 1101). For a 3600 RPM machine and an 80 toothed wheel, this frequency will be 4800 Hz. The signal should be approximately a sine wave (some distortion is normal) and have a magnitude of at least 5 volts PP (normally 30 to 40 volts PP will be obtained). Amplitude modulation of the signal will exist because of shaft run-out, imperfect toothed wheels, etc. This usually appears as a once per revolution frequency (60 Hz for a 3600 RPM machine). The existence of high frequency spikes usually indicates a shield grounding problem and a check should be made for the existence of more than one ground on the shield. Pull the Speed Transducer

OPERATION (continued)

INITIAL OPERATION (continued)

board (LOC 1101).

2. Turn on ± 15 VDC power using the toggle switch mounted on the faceplate of the Power Supply board (LOC 1111). On G6, which has no separate power supply, insure that the ± 15 VDC and 24 VDC busses are energized. Check that power supply voltages are correct. Adjustment of the ± 15 VDC supply can be made with board mounted rheostats. (Refer to theory of operation section.)

CAUTION

ON SOME EXCITATION EQUIPMENT, PULLING THE RELAY BOARD INTRODUCES A STEP INTO THE REGULATOR. THIS CAN RESULT IN DAMAGE TO THE GENERATOR. INSURE THAT THE REGULATOR IS IN THE MANUAL MODE OF OPERATION BEFORE PULLING THE RELAY BOARD.

3. If the generator is in the automatic mode of operation, transfer to manual. Pull the Relay board (LOC 1112). The Relay board is designed to interface with many excitation systems. Board mounted shorting clips and rheostats are used to adapt the relay board to the particular excitation system. (Refer to the drawings furnished with the equipment for the settings of F1S, F2S, F3S, F4S, F1P and F2P.) Check settings and readjust if necessary. Insert Relay board. Pull Washout and Output board (LOC 1108). Put shorting clip C5S on terminals 1, 2 (not GRD). Insert Extender board in LOC 1108).

CAUTION

THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO PLACE THE REGULATOR IN MANUAL MAY RESULT IN DAMAGE TO THE GENERATOR.

Insert a signal from pins 11, 12 to pins 3, 4 of the Extender board. This signal can be a dc signal or an oscillating signal (peak value should be less than 10 volts). With the PSS off (as evidenced by the "OFF" pushbutton being lighted) no voltage should exist at the test jack labeled "Out to Regulator" mounted on the Relay board. Pressing the "ON" pushbutton should cause the voltage to be present but attenuated for insertion into the appropriate regulator. If the "ON" pushbutton will not stay energized, then press the reset pushbutton mounted on the faceplate of the Ceiling Sensing board. If the "ON" pushbutton will still not stay energized, then check that the permissive contact is closed (usually 52G line breaker - see sheet 4 of Figure 4). If the unit is off line, then this permissive contact must be jumpered in order to establish that the output relay is working properly. Press the "OFF" pushbutton. The voltage at the test jack should be zero and the "OFF" pushbutton should be lighted. Remove Extender board. Insert Washout and Output board

OPERATION (continued)

INITIAL OPERATION (continued)

in LOC 1108. On Ceiling Sensing board (LOC 1110), put the delay dial on zero, the positive limit dial on 500 and the negative limit dial on 500. Press the reset pushbutton on the Ceiling Sensing board. Adjust the gain for 1.0 by turning the dial marked "Gain K" on the Washout and Output board to zero. Connect a variable source of dc voltage to "IN 1" of the Washout and Output board and set the voltage to some value less than 5 volts. Press the "ON" pushbutton on the Relay board. A voltage should exist at the test jack on the Relay board labeled "Out to Regulator". Increase the voltage at "IN1" of the Washout and Output board to ≈ 6 volts. After approximately 2.5 seconds, the FAR relay should drop out causing the "OFF" pushbutton to be lighted. Try to reinsert the PSS by pressing the "ON" pushbutton. The PSS should not latch in the on position. Remove the voltage from IN1. Press the reset pushbutton on the Ceiling Sensing board. Put the PSS in service (with the generator control in manual) by pressing the "ON" pushbutton. The PSS should latch in the on position and the "ON" pushbutton should be lighted. Remove the PSS from service by pressing the "OFF" pushbutton. The "OFF" pushbutton should be lighted. Some installations may have customer controls for the "PSS" mounted in the control room. Check that any external customer controls can be used to insert and remove the PSS. Leave the PSS in the off position.

CAUTION

THE FOLLOWING GAIN TEST MUST BE PERFORMED WITH THE PSS IN SERVICE AND THE REGULATOR IN THE AUTOMATIC MODE. INSURE THAT THE SHORTING CLIPS AND RHEOSTAT SETTINGS ON THE RELAY BOARD ARE CORRECT FOR THIS EQUIPMENT. INSURE THAT THE SPEED TRANSDUCER BOARD IS PULLED. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

The Relay board contains the necessary interface circuitry so that a one volt change in PSS output will cause a one percent change in generator open-circuit terminal voltage. With the PSS off, a small step of voltage (≈ 1 VDC) should be inserted into IN1 of the Washout and Output board. The regulator should be in the automatic mode of operation with the generator preferably off-line. The test may be satisfactorily performed on-line if the reactive current compensator (if any) is bypassed or switched to zero and if the voltage regulator of any other generator units paralleled at terminal voltage are temporarily turned to manual control. The gain is calculated by dividing the percent change in generator terminal voltage by the step in voltage applied to IN1. The gain should be (-) 1%/volt ± 0.1. The minus sign (terminal voltage decreases for a positive input voltage) arises because the signal is inserted into IN1 which bypasses the washout stage. F1P (on the Relay board) can be used to adjust the gain, if necessary.

An attempt has been made to interface with the voltage regulator in such a

OPERATION (continued)

INITIAL OPERATION (continued)

way that placing the PSS into and out of service will not cause a bump. On some equipments, F2P on the Relay board has been furnished for this purpose. With the voltage regulator in manual (or test if a test position is available on the control switch) pull the Speed Transducer board (LOC 1101). Place the PSS in service using the pushbuttons on the Relay board. Note whether the ac regulator output voltmeter or transfer voltmeter changes when the PSS is put in service. Use F2P on the Relay board to minimize the change in ac regulator output voltage when the PSS is inserted. Remove the voltage at IN1 and remove the PSS from service.

4. Check the operation of the Speed Transducer board (LOC 1101). Disconnect the leads from the toothed wheel/magnetic pickup at L23, L24. Connect an audio oscillator to L23, L24. Adjust the oscillator for approximately 10 volts peak-to-peak at 4800 Hz. The shorting clip on A1TB should be placed so that A1OC is in the circuit.

CAUTION

ON ALTERREX EXCITATION SYSTEMS, ANALYTICAL STUDIES AND FIELD TESTS INDICATE LACK OF TORSIONAL INTERACTION AND TORSIONAL SIGNAL FEEDTHROUGH WITH A1OC CONNECTED (5 Hz FILTER). FAILURE TO CONNECT A1OC COULD RESULT IN DAMAGE TO THE GENERATOR.

Insert the Speed Transducer board. The output of the Speed Transducer board should be zero (the dc level should be zero but there will be≈ 1 volt of 4800 Hz carrier feedthrough). If the voltage is more than ± 2.5 VDC, then readjust for 0 volts ± 0.2 volts. Increase the frequency of the audio oscillator to 5040 Hz. The transducer output should increase by + 5 volts from the 4800 Hz value (± 0.5 volts). Decrease the audio oscillator frequency to 4560 Hz and the transducer output should change by -5 volts from the 4800 Hz value (± 0.5 volts). The gain should not be adjusted unless the oscillator frequency calibration is accurate and the output is off by more than 0.5 volts. Remove the audio oscillator. Reconnect the leads from the toothed wheel/magnetic pickup to terminals L23, L24. With the turbine-generator synchronized, adjust null, A1P, for 0 volts ± 0.2 volts at the Speed Transducer output test jack. The Speed Transducer power supplies are isolated by 12 volt Zeners. If the null cannot be adjusted to 0 volts ± 0.2 volts, remove one of the shorting clips on the P1 terminal board mounted on the back of the rack. If the minimum voltage attainable is positive, then remove the shorting clip from P1-7 to P1-6. If the minimum voltage is negative, then remove the shorting clip from P1-5 to P1-6.

5. If it becomes necessary to check the frequency response of the PSS, set Signal Conditioning board (LOC 1102) for lead time constants of 0.5 seconds and lag time constants of 0.05 seconds. This can be accomplished by the following adjustments:

OPERATION (continued)

INITIAL OPERATION (continued)

TCT2V = 0 turns

TCT2F = 2.7 turns

TCT3 = 4.0 turns

TCT4 = 2.7 turns

TCT5 = 4.0 turns

On the Washout and Output board (LOC 1108) set the washout time constant for 10 seconds and the gain for 1.0. This can be accomplished by placing the shorting clip on C2S on terminals 1, 2, TCT1 dial on 5.4 turns and the gain control at zero. The twin tee notch filter on the Signal Conditioning board should be adjusted for the proper frequency (60 Hz for a 2-pole machine and 30 Hz for a 4-pole machine). A 30 Hz notch filter should consist of B16C through B23C and associated resistors and rheostats. For a 60 Hz notch filter, capacitors B20C through B23C are removed.

CAUTION

THE PSS MUST BE OUT OF SERVICE FOR THE FREQUENCY RESPONSE TESTS. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

Pull the Speed Transducer board (LOC 1101). Insert the Extender board in place of the Speed Transducer board. Connect an audio oscillator to terminal 15 or 16 and ground terminal 3 or 4 of the Extender board. Measure the frequency response of the PSS from 0.1 Hz to 100 Hz, measuring oscillator output and output test connector on the Washout and Output board.

The oscillator output level should be held at 100 Mv zero to peak (200 Mv PP) to avoid saturation of the amplifiers. Compare gain and phase to values shown in Figure 1.

6. Check out the Ceiling Sensing board (LOC 1110). Pull the Ceiling Sensing board and verify that the shorting clip on E1S is between 1 and 2. Reinsert the board. Set both the positive limit (E2P) and negative limit (E3P) for 500 (5.0 turns). Set the time delay on the Ceiling Sensing board for minimum (0). Insert a variable dc voltage into IN1 of the Washout and Output board. Place the regulator in the manual mode of operation.

CAUTION

THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO DO THIS COULD RESULT IN DAMAGE TO THE GENERATOR.

OPERATION (continued)

INITIAL OPERATION (continued)

Place the PSS in service by pressing the "ON" pushbutton on the Relay board. Slowly increase the voltage until the PSS trips out of service as evidenced by the "OFF" pushbutton being lighted. The voltage at Out 1 on the Washout and Output board should be 5 volts ± 0.25 volts. Remove the voltage. Pressing the "ON" pushbutton on the Relay board should have no effect. Press the reset pushbutton on the Ceiling Sensing board. Pressing the "ON" pushbutton on the Relay board should now cause the PSS to be placed in service. Reverse the voltage into IN1 of the Washout and Output board. The same procedure can be followed to verify that this part of the circuit is working. Adjust the time delay (EIP) on the Ceiling Sensing board for 300 (3 turns). Remove the voltage at IN1 of the Washout and Output board and press the reset pushbutton on the Ceiling Sensing board. Place the PSS in service (regulator in manual). Put a voltage into IN1 of sufficient magnitude to cause a trip. The PSS should trip to "OFF" approximately 10 seconds after insertion of the signal. Remove the test voltage and reset the Ceiling Sensing board by pressing the reset pushbutton on the faceplate. Place the positive and negative limit dials on 1000 (10 turns) and the time delay dial on zero.

7. On Washout and Output board, set negative limit dial and positive limit dial on 500 (5 turns). Insert a variable dc voltage into IN1. Monitor the output voltage at OUT1 using an oscilloscope or voltmeter. Increase the dc voltage slowly and note at what voltage the output fails to follow the input voltage (output voltage is limited). This voltage should be 4.75 volts ± 0.5 volts. Reverse the voltage and perform the same test. The output voltage should be of the opposite polarity but same magnitude (within ± 0.5 volts). Remove the voltage source and Extender board. Reinsert the Speed Transducer board. Insure that the PSS is off. Place the regulator in the automatic mode of operation.

8. The adjustment range provided by the PSS can result in extremely high gain over a range of frequencies. For instance, with K set at 100, T2 and T3 set at 1.5 sec and 0.02 sec respectively, the gain will be 116 Db (631,000). Although this gain will probably never be used in an actual system, a good check of PSS noise level can be obtained using this gain. Insure that the PSS is off. On the Signal Conditioning board (LOC 1102), set TCT2F and TCT4 for 1000 (10 turns) and TCT3 and TCT5 for zero.

   Pull the Washout and Output board (LOC 1108). Put the shorting clip on C5S on terminals 2, 3 (GRD). Put the shorting clip on C3S on terminals 2, 3. Insert Washout and Output board. Turn gain adjustment to 1000 (10 turns). Remove Speed Transducer board (LOC 1101). Insert Extender board in place of the Speed Transducer. Short from pin 15 to pin 3 of the Extender board. Using an oscilloscope with a very slow sweep speed (1 sec/cm), determine the peak-to-peak output voltage (measured at OUT 1 of the Washout and Output board). This voltage should be observed for some time, perhaps 10 seconds. This voltage should be less than 3 volts to 1 minute peak-to-peak. Remove Extender board. Insert Speed Transducer board in LOC 1101. Remove Washout and Output board. Place C5S shorting clip on terminals 1, 2 (not GRD). Place

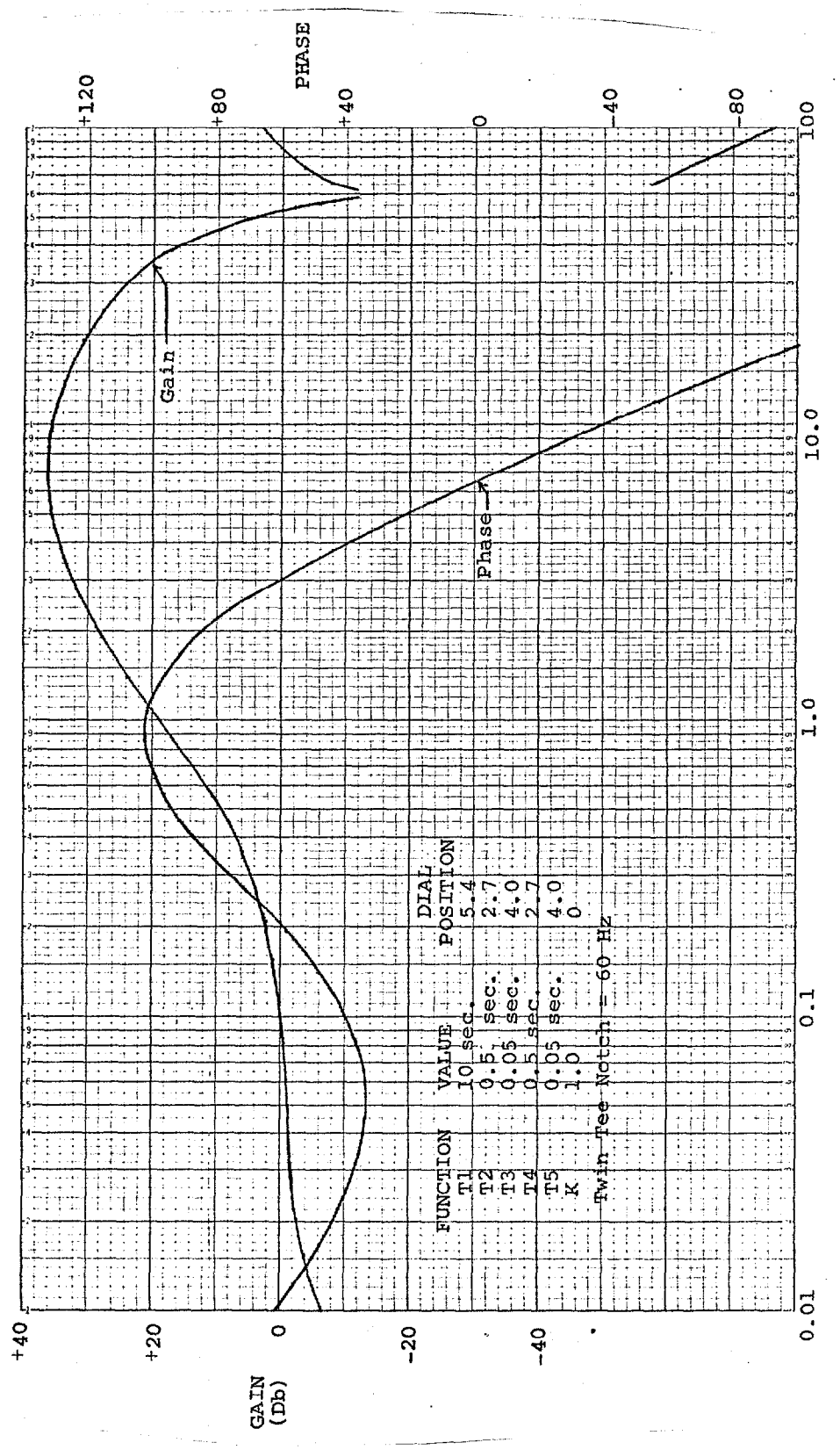

Gain and Phase Vs. Frequency
for PSS (Excluding Input Transducer)
PHASE
+120
+80
+40
0
-40
-80
Gain
Phase
DIAL
FUNCTION VALUE POSITION
T1 10 sec. 5.4
T2 0.5 sec. 2.7
T3 0.05 sec. 4.0
T4 0.5 sec. 2.7
T5 0.05 sec. 4.0
K 1.0 0
Twin Tee Notch = 60 Hz
GAIN (Db)
+40
+20
0
-20
-40
0.01
0.1
1.0
10.0
100
FREQUENCY (HZ)

Figure 1

OPERATION (continued)

INITIAL OPERATION (continued)

C3S shorting clip on terminals 1, 2. Return gain adjustment dial to zero (unity gain). Insert Washout and Output board. Final settings of time constants and gain can only be made after on-line tests and/or system studies.

9. Check out the following optional circuits if supplied with this equipment.

A. Check out the Voltage Sensing panel located below the printed circuit board rack. Remove the Voltage Deviation Sensing board and the Under-Overvoltage Detection board. Apply 120 VAC to L25, L26 and L27 (ØA, ØB and ØC respectively). Close the isolating switches mounted on the "L" terminal board at L25, L26, and L27. The voltage from LB5-B to LB5-A should be 38 VDC ± 10%. Replace the Voltage Deviation Sensing and Under-Overvoltage Detection boards.

B. Check out the Under-Overvoltage Detection board (LOC 1109). Put the Under-Overvoltage Detection board on an Extender board. Measure the voltage at D5TP. If the voltage is not +5 VDC ± 0.05 VDC, then adjust using D4P. Measure the voltage at D6TP. If the voltage is not -5 VDC ± 0.05 VDC, then adjust using D6P. Measure the voltage at D2TP. If the voltage is not -5 VDC ± 0.05 VDC, then adjust using D2P. At an input voltage corresponding to rated, measure and record the voltage at D1TP. Adjust D1P (input voltage) for zero volts at OUT 1. Place C1S shorting clip on the Washout and Input board on terminals 1, 2. The voltage at OUT 2 should be > 2.0 VDC. Adjust the positive and negative limit dial to 0 (0 turns). Change the generator voltage by approximately 1% (or change the voltage at pins 15, 16 by 1% if a dummy voltage source is used). Turn the appropriate limit dial (if voltage was raised, use the positive limit and if lowered, use the negative limit) and determine the setting at which OUT 2 changes to approximately 0.5 volts. The dial setting should be 100 ± 5. When terminal voltage is changed back to nominal, the voltage at OUT 2 should return to > 2.0 VDC.

With the C5S shorting switch on terminals 1, 2 (on the Washout and Output board), insert a +1 VDC signal into IN1 of the Washout and Output board. If noise is present in the output, it may be necessary to pull the Signal Conditioning board. With the gain dial set for zero, the voltage at OUT 1 on the Washout and Output board should be -1 VDC. Change terminal voltage by an amount sufficient to cause the voltage at OUT 2 of the Under-Overvoltage Detection board to change to approximately 0.5 VDC. The voltage at OUT 1 on the Washout and Output board should be reduced to zero volts (± 0.1 VDC). Change terminal voltage back to nominal and the voltage at OUT 1 on the Washout and Output board should return to -1 VDC after approximately 0.5 seconds. Remove the signal from IN1 of the Signal Conditioning board. The positive and negative limit dials should be set to the values specified by the customer. C5S should be left on terminals 2 and 3 and C1S should be left on terminals 1 and 2.

C. Check out the Voltage Deviation Sensing board (LOC 1107). Put the Voltage

OPERATION (continued)

INITIAL OPERATION (continued)

Deviation Sensing board on the Extender board. Adjust the gain dial for 1000. Place B1S (Signal Conditioning board) on terminals 2, 3. Measure the voltage at H1TP. If this voltage is not -5 VDC ± 0.05 VDC, then adjust using H2P. Place H1S shorting switch on terminals 2, 3 (GRD). With rated generator terminal voltage (or with a source of dc voltage connected from pin 11 to pin 13 and adjusted for approximately 30 VDC) adjust H1P for zero volts at OUT. Change terminal voltage by 1% (or the dc source by 1%). The voltage at OUT should be 0.26 ± 0.025 VDC. If an external source is used to simulate a terminal voltage signal, the signal should be inserted between pins 11, 12 and pins 13, 14. Open the PT isolation switches at L25, L26 and L27 if an external source is used. Note that the Voltage Deviation Sensing board causes an inversion of the signal. Return the gain dial to zero, remove the external voltage source (if used) and close the PT isolation switches. Remove the Extender board and reinsert the Voltage Deviation Sensing board. B1S should be left on terminals 1 and 2 and H1S should be left on terminals 2 and 3.

Initial Settings

The PSS has been designed to interface with various excitation systems. In order to permit compensation for widely different systems, adjustments are made using calibrated dials. On the Washout and Output board, set the gain dial to zero and C3S on terminals 1 and 2. Insure that the Relay board has the correct rheostat and switch settings. Check the Speed Transducer board to insure that A1OC is in the circuit (5 Hz filter). If the optional Under-Overvoltage Detection board is being used, insure that C1S (Washout and Output board) has the shorting link on terminals 1, 2. If the optional Voltage Deviation Sensing board is being used, insure that B1S (Signal Conditioning board) has the shorting link on terminals 1, 2.

Settings for a particular installation are best obtained by analytical studies and/or on-line tests. It is the responsibility of the customer to perform the analytical studies and on-line tests necessary to determine proper settings. General Electric Company can provide assistance in performing analytical studies and system tests. Contact your local sales office. A paper, "Concepts of Synchronous Machine Stability as Affected by Excitation Control", by F. P. deMello and C. Concordia may be used as a reference (68-TP-129-PWR). Other publications which may be used as references are "Test Procedure for Power System Stabilizers" published by the Western Systems Coordinating Council and an IEEE publication "Guide for the Identification, Testing and Evaluation of the Dynamic Performance of Excitation Control Systems" paragraph 6.03 (P421, 1977).

On-Line Tests

Power System Stabilizers are employed to provide an additional input to the regulator to improve power system dynamic performance. They have been employed to solve two separate but related problems; local machine-system oscillations in the 1-2 Hz range and inter-area oscillations $\leq$ 0.5 Hz. The fundamental function of the signal conditioning network of the PSS is to compensate for the phase lags of the

OPERATION (continued)

INITIAL OPERATION (continued)

system being controlled. For best contribution to system damping, the gain of the PSS should be maximized under the most sensitive conditions (usually at or near full-load). The desired stabilizing signal has as its reference the base system frequency which is slowly changing all the time. The washout function serves to minimize the effect of overall system speed changes on machine terminal voltage.

CAUTION

THE PSS HAS A WIDE RANGE OF ADJUSTMENT OF LEAD AND LAG TIME CONSTANTS. DO NOT ADJUST FOR A DIFFERENCE IN LEAD-LAG SETTINGS OF MORE THAN TEN TO ONE. LARGER DIFFERENCES IN SETTINGS CAN CAUSE EXCESSIVE AMPLIFICATION OF TORSIONAL SIGNALS AND RESULT IN DAMAGE TO THE TURBINE-GENERATOR.

The PSS has adjustable limits on the output signal so that the PSS will not completely override voltage regulator action. The output voltage limits are independently adjustable but should be set identically in order to prevent nonsymmetrical limiting which might bias the voltage regulator. Since the ceiling sensing function occurs after the adjustable output limit stage, the output voltage limits should be set at a higher voltage than the ceiling sensing limits.

In order to facilitate on-line frequency response testing of the excitation system, IN1 is provided on the Washout and Output board. With the shorting clip on C5S on terminals 1, 2, a drive signal can be inserted into IN1 which will produce a calibrated terminal voltage change through the regulator. For normal operation, the shorting clip on C5S should be on terminals 2, 3 (GRD) and the shorting clip on C1S should be on terminals 2, 3.

Figure 2 is provided so that dial settings can be chosen to realize the desired time constants and limit settings.

PRINCIPLES OF OPERATION

GENERAL

The Power System Stabilizer (PSS) is used to provide an additional input to the voltage regulator which, under certain system conditions, can improve power system dynamic stability. A signal is obtained from a toothed wheel/magnetic pickup mounted at the turbine front standard. This signal is frequency modulated with the carrier frequency being 4800 Hz (for a 3600 RPM machine and 80 toothed wheel) and the modulation frequency being equal to the speed changes of the turbine shaft. This frequency modulated signal is passed through a speed transducer to produce an output signal proportional to the magnitude of the deviation in speed with a frequency equal to the speed deviation frequency. This demodulated signal is then passed through two stages of lead-lag compensation to provide phase com-

PARAMETER VALUE VS DIAL SETTING

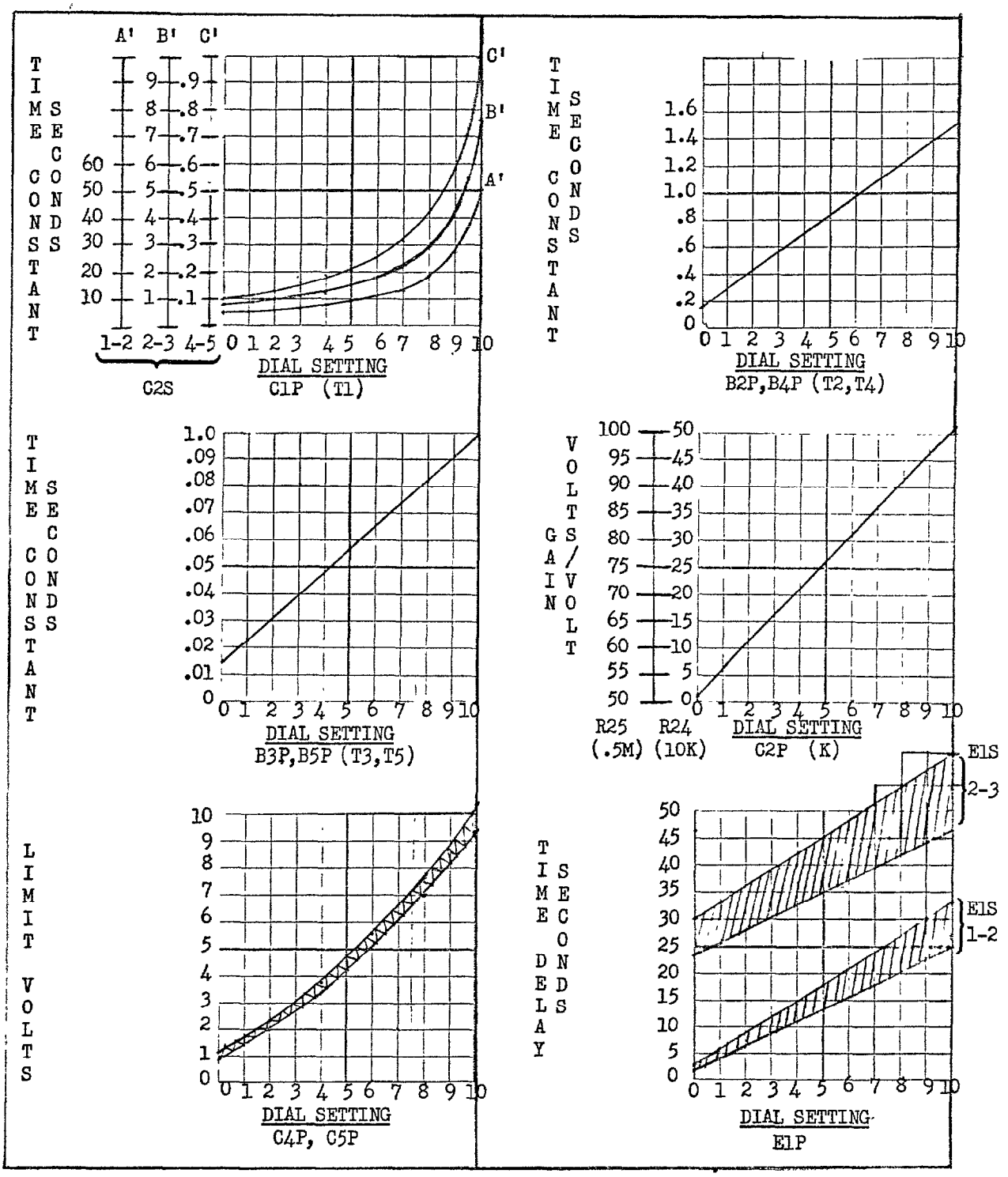


Figure 2

PRINCIPLES OF OPERATION (continued)

GENERAL (continued)

pensation for the system, a twin-tee notch filter tuned to 60 Hz (30 Hz for an 1800 RPM machine), a washout stage to remove drift and slow system frequency changes and an adjustable gain stage to maximize the effect of the PSS. Limits are provided on the output stage so that during large system disturbances, the PSS will not interfere with voltage regulator action. A relay is provided for placing the PSS in service (and also removal from service). Interfacing is provided to properly scale the PSS output signal so that 1 volt from the output stage produces a 1% change in terminal voltage. Protective circuits are provided to detect a failure which causes the output to go to ceiling (Ceiling Sensing board) and, optionally, a failure or system condition which causes terminal voltage to deviate by more than a preset value from nominal (Under-Overvoltage Detection board). Ten-turn rheostats with calibrated dials are furnished for all customer adjustments. Figure 4 contains an overall block diagram of the PSS. Individual board and panel elementary diagrams are furnished with the equipment. It will be necessary to have these for much of the following explanations.

POWER SUPPLY SYSTEM

The rack-mounted PSS (G6) utilizes the power supply of the cubicle in which it is mounted. Refer to the instruction manual for the voltage regulator cubicle for a description of the power supply.

The panel mounted PSS (G5) includes both the ± 15 VDC and 24 VDC power supplies. The 115 VAC, 60 Hz power should be connected to L21 and L22. The S1T and S2T transform the input voltage (and provide isolation) and then the voltage is rectified by a single phase full-wave bridge, filtered to remove the dc component, limited by Zener diode action and then fed to series pass regulators. Fuses R1F and R2F, SCR's R1CR and R2CR, along with associated sensing and firing circuits form a "crowbar" circuit which will protect sensitive semiconductor components from overvoltage by blowing a fuse and thus removing the excess voltage. IC voltage regulators (G1A, G2A) are used to control the output voltage. The G2P and G4P mounted on the printed circuit board provide adjustment of the output voltage. The G1R, G1P, G2R, G3R provide a current limit feature on the +15 volt regulator (a similar circuit is found on the -15 volt regulator). The current limit is of the foldback type, i.e., the maximum output current is reduced as the load on the regulator is increased above the full-load value. This insures that critical components in the regulator (series pass transistors) will not be damaged by a short circuit. The current limit can be adjusted using G1P and G5P. The positive limit is factory set at 1 Amp and the negative limit at 750 Ma (these are not the short circuit current values). Integrated circuits J1A and J2A provide voltage sensing for the "crowbar" circuit. The J1P and J2P are factory adjusted to cause the "crowbar" circuit to operate at a supply voltage of 17 volts. The S1T and S2T have taps which will permit variations in the input voltage from the nominal 115 VAC value. (Refer to the diagram for the correct tap for a given input voltage.) In order to reduce the noise level to as low a value as possible, the ± 15 volt power supply is very tightly regulated and heavily filtered. The regulation is 0.1% and the ripple on the power supply should be ≤ 5 Mv PP. The 24 volt supply does not require tight regulation since

PRINCIPLES OF OPERATION (continued)

POWER SUPPLY SYSTEM (continued)

it is used to drive relays and energize lights. The S3T is used to isolate the 24 volt supply from the ± 15 volt supply and to transform the input voltage to a lower voltage. This voltage is rectified by a full-wave bridge and then filtered. Regulation is provided by a Zener diode (S1Z). The 24 volt power supply is ungrounded while the ± 15 volt supplies are grounded through a parallel resistor capacitor combination. The high impedance ground on the ± 15 volt supplies allows the PSS to be used with both grounded and ungrounded systems without ground loop problems.

SPEED TRANSDUCER BOARD

The input signal is transformer coupled through 1T. The 1Q is used as a zero crossing detector and with 2Q forms a high gain stage so that whenever a zero crossing is detected, the voltage at 4TP will increase abruptly to 15 volts. This sudden transition in voltage is capacitively coupled through 1C to a monostable multivibrator composed of 3Q, 4Q and associated circuitry. A pulse is generated at 6TP that has an off-time determined by 2C and 8R. During the time that the input signal is negative, 1D protects 1Q and the monostable multivibrator is not triggered. 6TP stays at -5 volts. When the input voltage goes positive by approximately 1 diode drop (VBE), a pulse is generated which triggers the monostable multivibrator and the voltage at 6TP goes to approximately -15 volts. This pulse lasts for a period of time determined by 2C and 8R. The average value of the signal at 6TP is thus determined by the ratio of the on time to off time (on is -5 volts, off is -15 volts). The off time is fixed by 2C and 8R, therefore, the average voltage at 6TP will vary with input frequency. The signal at 6TP is fed through a low pass filter to summing amplifier 1A. A signal is taken from 1P to null the output of 1A at rated frequency so that deviations in frequency are detected. The 2P and 17R are used as feedback for 1A. The 2P is adjusted to give a 1 volt change for a 1% change in input frequency. Provisions have been made so that the feedback time constant can be selected for 280 Hz or 5 Hz.

SIGNAL CONDITIONING BOARD

The Signal Conditioning board consists of two lead-lag stages and a twin-tee notch filter. In addition to the signal from the Speed Transducer, an extra input with separate lead time constant adjustment is provided for a voltage input (IN2). The B1P provides adjustment of the first stage lead time constant, which is proportional to B1C, B1R and the setting of B1P. The B3P provides adjustment of the lag time constant which is proportional to B9R, B4C and the setting of B3P. The dc gain of the stage is B9R/B1R which is 1.0. The B3R, B3C, and the equivalent resistance of B3P cause extraneous lead and lag time constants which have been minimized but may be significant when performing calculations at torsional frequencies. The second lead-lag stage is identical to the first except for the addition of an output voltage limit circuit formed by B1Z, B2Z, B5D, B6D and a drive circuit formed by B1Q, B21R, B22R, B3D and B4D to increase the output current capability of B2A. A twin-tee notch filter is placed between the first and second lead-lag stages to remove 60 Hz (for 3600 RPM machines) or 30 Hz. (for

PRINCIPLES OF OPERATION (continued)

SIGNAL CONDITIONING BOARD (continued)

1800 RPM machines). The twin-tee is factory adjusted to produce greater than 50 Db attenuation at the notch frequency. The notch frequency is given by

$$F = \frac{1}{2\pi\ (B11R+B4P)\ (B16C)}$$

WASHOUT AND OUTPUT BOARD

The Washout and Output board contains a washout function, a gain function and a power amplification function. Operational amplifier C1A is a chopper stabilized device for low offset. C1A is basically a derivative circuit formed by C1C and the feedback resistor (C2R, C3R or C4R). C2C is added to make the derivative circuit only effective for low frequencies and C1P is used to provide adjustment of the effective feedback resistance and thus control the time constant. Coarse adjustment of the time constant is available through C2S. The C9D, C10D, Z1C, Z2C, C31R and C32R form a clamp to limit the output voltage of the washout stage. An extra input is available at IN1 which bypasses the washout stage. This allows insertion of a signal into the voltage regulator for calibration of the PSS or for frequency response testing of the system. The C5S is used to ground IN1 during normal use. Changing C5S shorting clip to terminals 1, 2 allows IN1 to be used for driving the regulator. The C2A is used as an adjustable gain stage with the gain being determined by C3S and C2P. The C3 is used to select a range of gain from 1 to 50 or 50 to 100. An adjustable limit is provided on the output of C2A by C1Q, C2Q, C15R, C16R, C1D, C2D, C4P, C5P, C13R and C14R. The output of C2A is used as the input signal for Ceiling Sensing board through pins 13, 14. The C3Q and C4Q are junction FET's used as a static switch. The IN3 can be connected to an optional Under-Overvoltage Detection board which serves to disconnect the PSS whenever terminal voltage deviates by an adjustable amount. In normal operation, C1S should have the shorting clip between terminals 2, 3 (OUT). The C3Q will be conducting and C4Q will be non-conducting. The C3A is connected to a push-pull transistor stage (C7Q, C8Q, C9Q, C10Q) in order to increase the output current capability to ± 100 Ma. The C3A and the push-pull stage are connected as a unity gain voltage follower.

RELAY BOARD

The Relay board is used to control insertion of the PSS into the regulator, provide the proper interfacing with different voltage regulators and provide interfacing with customer equipment. Meter M1 is mounted on the faceplate and can be used to insure that the PSS output is zero before insertion into the voltage regulator. "OFF" and "ON" pushbuttons with lights are mounted on the faceplate and can be used to insert or remove the PSS from service. Lights mounted in the pushbutton indicate the status of the PSS. A line breaker contact (52G) should be used as a permissive contact so that the PSS is inoperative off-line. The pushbuttons have momentary contacts and so F93 is sealed in by an FAR and F93 contact. The F93A is an auxiliary relay used to increase the number of contacts of F93. FAR is controlled by the Ceiling Sensing board.

PRINCIPLES OF OPERATION (continued)

RELAY BOARD (continued)

The F1R, F2R, and F1P are used to properly scale the PSS signal for insertion into the regulator. The F2P is used to dummy load the regulator so that a bump in excitation will not occur whenever the PSS is placed in service. (Refer to the diagram for the proper settings of F1P and F2P and switches F1S, F2S, F3S, F4S.)

CEILING SENSING BOARD

The Ceiling Sensing board is used to detect a failure in the PSS which causes the output to be greater than a specified value for an adjustable length of time. Once the condition exists, the PSS cannot be restored to service until a reset pushbutton is depressed on the Ceiling Sensing board.

The E1A and E2A operate as comparators with E2P and E3P used to adjust the level at which a failure is detected. If the input voltage is less than the positive limit (E2P) and more positive than the negative limit (E3P) then the voltage at E6TP is at a minimum. The voltage at E5TP is passed through an inverter in E1L and then a nand operation is performed on the signal from E6TP and the inverted signal from E5TP. A nand gate has its output in the high state unless all inputs are in the high state.

In this case, both inputs are in the high state so its output is in the low state. The output of this nand gate is passed through two inverters (no change in state) and is also used to trigger a one-shot (E3L). With a ceiling condition not detected, the one-shot trigger input will stay in the low state and the output (the complemented output is used in this case) stays in the high state. The inputs to the first nand gate of E2L are a low and a high, which means the output will be in the high state. Two of the nand gates are used in a cross-coupled fashion to allow latching whenever a ceiling condition is detected. The E8C is provided to insure that the cross-coupled gates will go to the proper state upon energization. With no ceiling condition detected, the inverter output of E2L will be in the low state, turning on transistors E2Q and E3Q and energizing relay FAR on the Relay board.

If a ceiling condition exists (whether positive or negative), the second nand gate of E1L will have its input states changed to one low and one high. The output of this nand gate will go to the high state which triggers the one-shot (E3L) and changes the input on the first nand gate of E2L to a high state. Nothing happens however, because the output of the one-shot (E3L) is in the low state and remains there for a period of time determined by E7C, E1P and E8R or E9R (depending on the setting of E1S). After the timing interval, the output of the one-shot (E3L) goes to the high state and the output of the first nand gate of E2L goes to the low state. The output of the cross-coupled gates of E2L goes to the low state and will remain low even if the ceiling condition is cleared. The inverter output of E2L changes to the high state, which turns off E2Q and E3Q and drops out relay FAR on the Relay board, giving an alarm. To clear the alarm and place the PSS back in service, it is necessary to press the reset pushbutton (E1PB). This changes the input to one of the cross-coupled gates from a high to a low state which forces the output of the inverter to be in a low state. The cross-coupled gates latch in this state, the relay FAR is picked up and the alarm is cleared. Table 1 illustrates the logic levels at the various board test points for both normal and abnormal input signals.

PRINCIPLES OF OPERATION (continued)

CEILING SENSING BOARD (continued)

Input Voltage	E5TP	E6TP	E7TP	E8TP Init.	E8TP Time Delay	E9TP	11TP and "Out" Init.	11TP and "Out" Time Delay
Normal	0	1	0	1		0	0	
High Pos.	1	1	1	0	→ 1	1	0	→ 1
High Neg.	0	0	1	0	→ 1	1	0	→ 1

Note: A zero is < 0.6 volts
A one is > 2 volts

Table 1

Logic Levels for Ceiling Sensing Board

OPTIONAL EQUIPMENT

Under-Overvoltage Detection Board (Optional)

The Under-Overvoltage Detection board (LOC 1109) is used to remove the PSS from service whenever generator terminal voltage deviates by a preset amount from nominal. The D2P is used to establish a reference for the input voltage signal and insure proper scaling. The D1P is used to null out the reference voltage at rated terminal voltage so that a change from rated is detected. A faceplate mounted test point (IN2) is provided for insertion of a test signal. The D2A and D3A are used as comparators with D3P and D4P establishing a variable reference point. The D4P and D6P provide proper scaling so that D3P and D4P dial setting (0 to 10) corresponds to percent voltage deviation from nominal. The output normally stays at > 2.0 VDC with no undervoltage or overvoltage detected. When an undervoltage or overvoltage is detected, the output goes to< 0.5 VDC. If terminal voltage changes back to a value less than the limits, the output changes back to > 2 VDC after 1/2 second time delay. The output of the Under-Overvoltage Detection board is used to control a static switch on the Washout and Output board. If the Under-Overvoltage Detection board is to be used, then the shorting link on C1S must be placed between terminals 1 and 2. Nominal input voltage is 30 VDC at rated terminal voltage.

Voltage Deviation Sensing Board (Optional)

The Voltage Deviation Sensing board (LOC 1107) is used to provide an input to the PSS proportional to the deviation of terminal voltage from rated. The input

PRINCIPLES OF OPERATION (continued)

OPTIONAL EQUIPMENT (continued)

signal is obtained from the Voltage Sensing panel and is nominally 30 VDC at rated generator terminal voltage. The H2P is used to provide a reference voltage and to allow proper scaling of the input signal. The H1P is used to null out the reference voltage at rated terminal voltage. A faceplate mounted test point (IN2) allows insertion of a signal for test purposes. Gain adjustment is provided for a range of 0.2 to 5.2 volts per volt or 0.01 to 0.26 per unit/per unit. If the Voltage Deviation Sensing board is to be used, the shorting link on B1S (Signal Conditioning board) must be on terminals 2 and 3, (not GRD).

MAINTENANCE

The PSS is static having no moving parts except for relays and instruments, therefore, little maintenance should be required. If excessive vibration is present, all screw connections should be checked periodically. The Speed Transducer null adjustment should be checked periodically and readjusted if necessary. A pushbutton is furnished on the Speed Transducer which puts a small step into the Speed Transducer output. This step is equivalent to approximately 0.030% change in speed. This change is so small that system noise may mask the effect on some systems. If a response can be observed in the output of the Washout and Output board when pressing the pushbutton, record for future reference. A quick check on the operating status of the PSS can then be made by pressing the pushbutton and comparing the output with the recorded output. The following photograph (Figure 3) shows the output of the Washout and Output board (T2 = T4 = 0.5 sec, T3 = T5 = 0.05 sec, T1 = 10 sec and K = 1.0) when the pushbutton is pressed and then released.

TROUBLESHOOTING

1. Carefully check all wiring if problems are experienced at start-up.

2. If no response is obtained while making the frequency response test:

   a. Check that the power switch on the Power Supply board is on (if a Power Supply board is furnished).

   b. Check fuses S1F and S2F on the Power Supply panel.

   c. Check that all boards are fully inserted.

   d. Check for proper jumper connections on Washout and Output board.

   e. Check each board individually for a response.

3. If an incorrect signal is obtained in terminal voltage, check the switch and rheostat settings on the Relay board.



PSS JOG TEST

Vert. = 500 mv/div
Horiz. = 200 msec/div

Figure 3

TROUBLESHOOTING (continued)

4. If the PSS will not stay in service:

   a. Reset Ceiling Sensing

   b. Check that permissive contact is closed.

5. Speed Transducer output stays at ceiling:

   a. Insure that generator speed is at rated.

   b. Readjust null.

   c. Remove appropriate shorting clip on P1 terminal board (see Speed Transducer Initial Operation section).

6. An instability is observed with gain dial at zero:

   a. Measure gain to ensure that it is unity.

   b. Insure that an increase in frequency causes an increase in generator field voltage.

7. If a board is found to be defective, replace it and return the defective board for repair and return.

RENEWAL PARTS

When ordering renewal parts, the following information should be given:

1. Catalog number stamped on the part, with complete description including use and location. Board components should not be replaced; order the entire board.

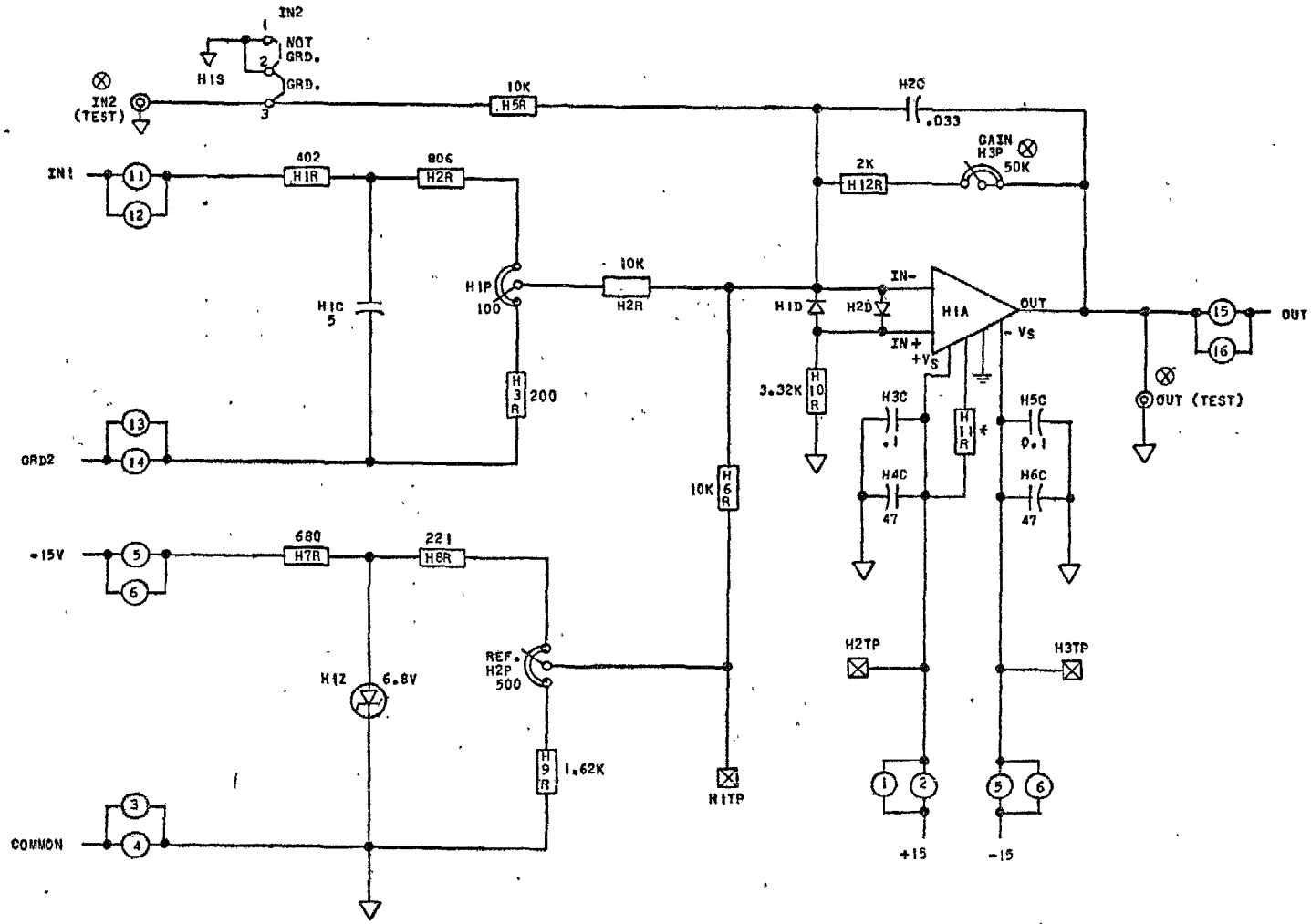
2. Complete nameplate data appearing on the assembly that included the component, including the serial number of the assembly.

3. If possible, data on original order on which equipment was first supplied, including all numerical references.

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
VOLTAGE DEVIATION SENSING
POWER SYSTEM STABILIZER
FIRST MADE FOR 397932MA324G1, LA202G1
ELEMENTARY DIAGRAM



NOTES:

⊗ - MOUNTED ON FACEPLATE

RESISTOR VALUES ARE IN Ω AND ARE 1/2 WATT.
CAPACITOR VALUES ARE IN MFD.

Figure 5

REVISIONS

PRINTS TO 11

MADE BY IRV MAYER 4/7/73
ISSUED S. Mayer May 18, 1973
APPROVALS T. Smith
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C306564

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE POWER SYSTEM STABILIZER

FIRST MADE FOR 3S7932LA202G5

FUNCTIONAL ELEMENTARY

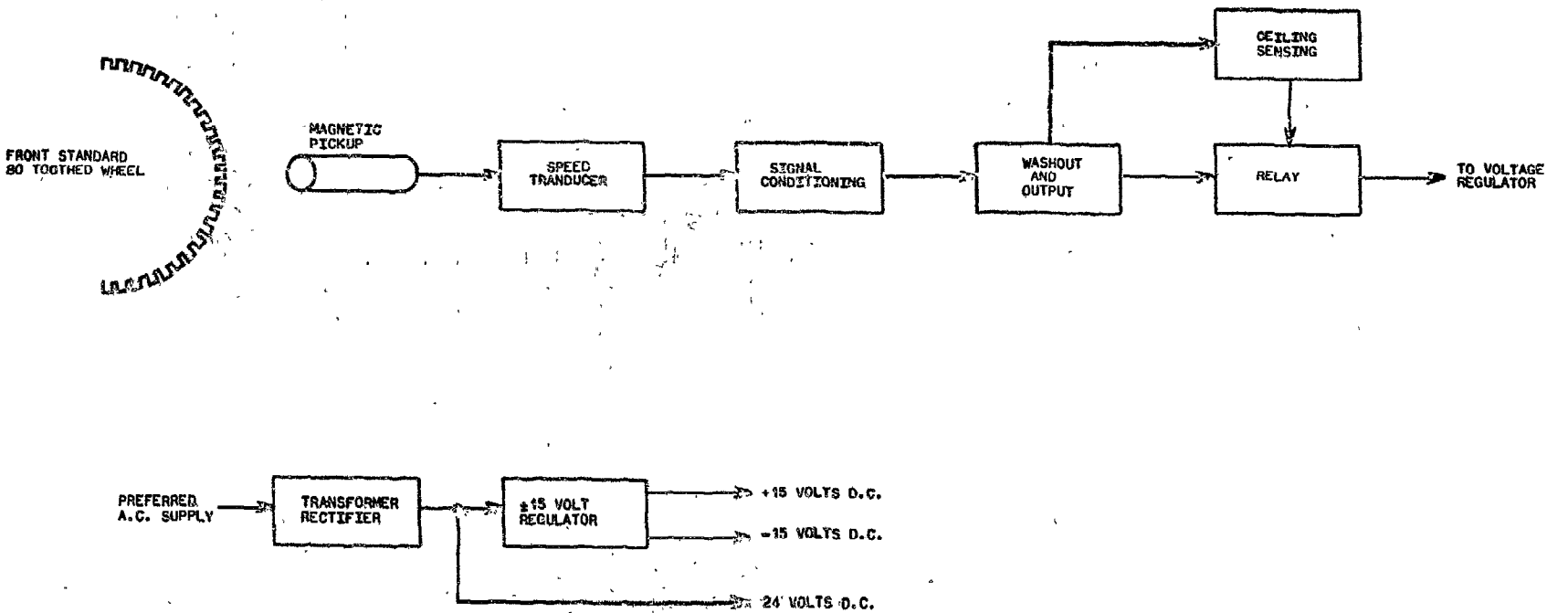


Figure 4, Sheet 1

REVISIONS

PRINTS TO 711

ISSUED OCT 2 2 1979

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

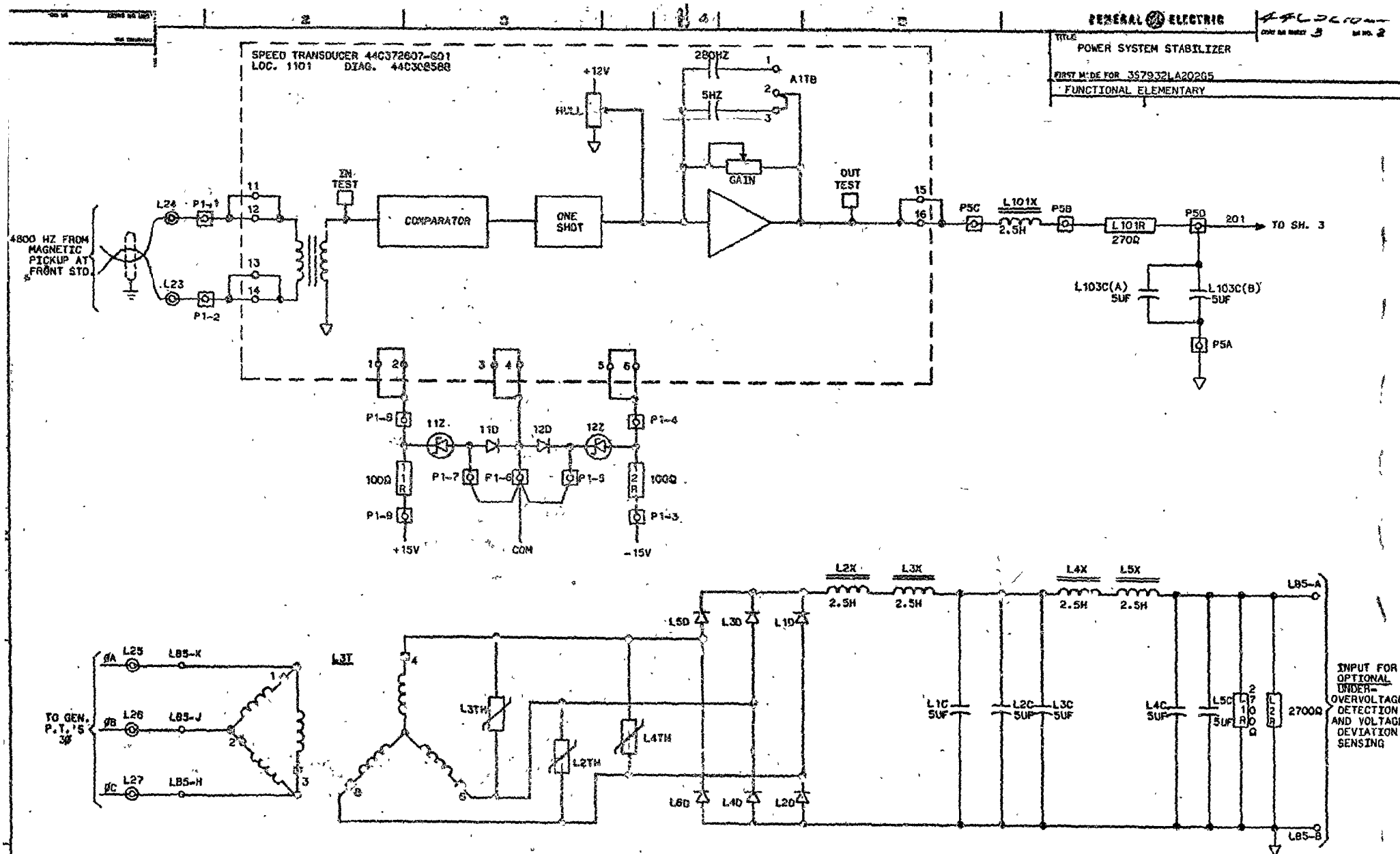


Figure 4, Sheet 2

REVISIONS
1 DEC 1 1976
2 DEC 2 2 1976
3 DEC 21 1977
PRINTS TO
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

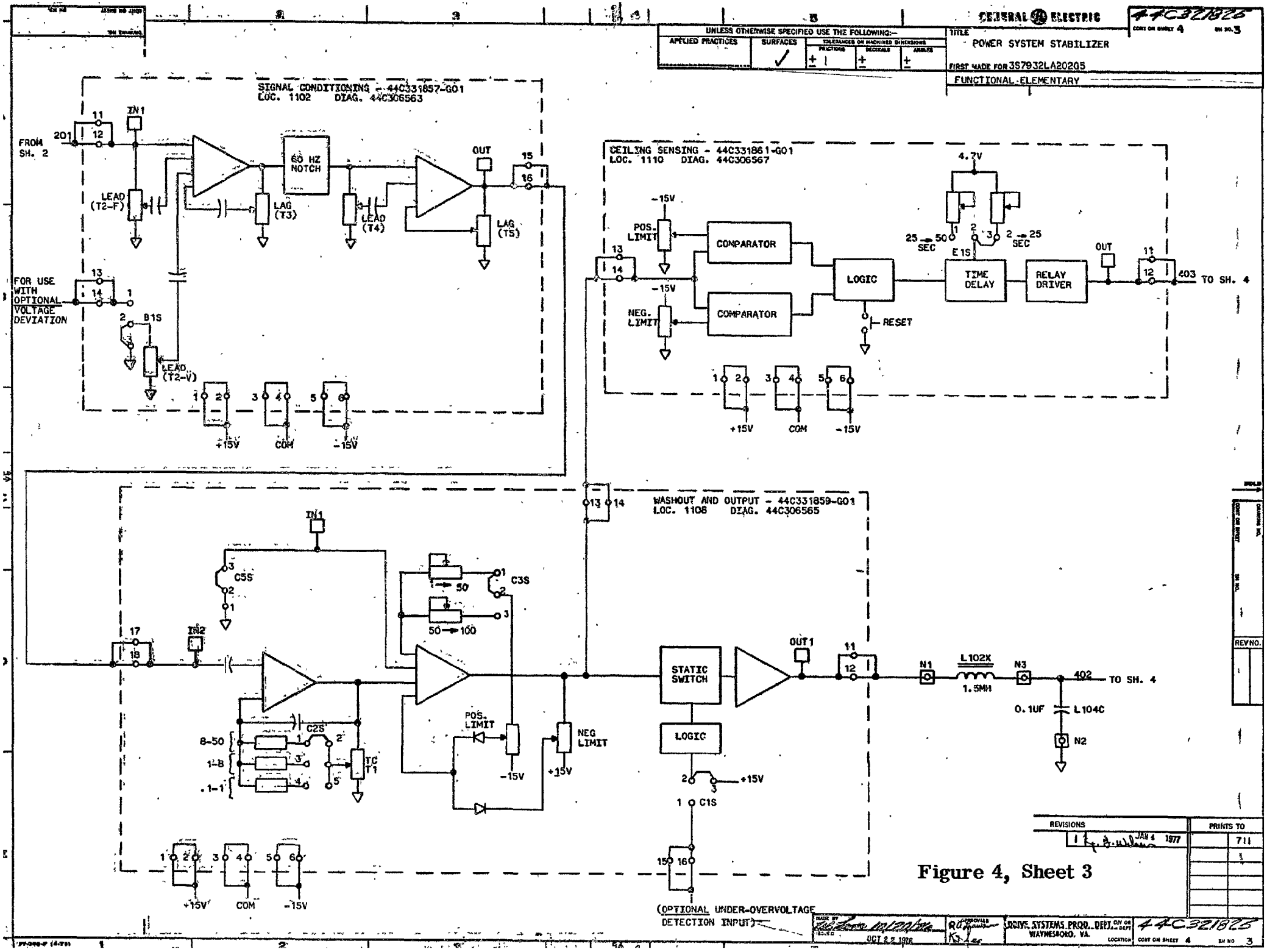
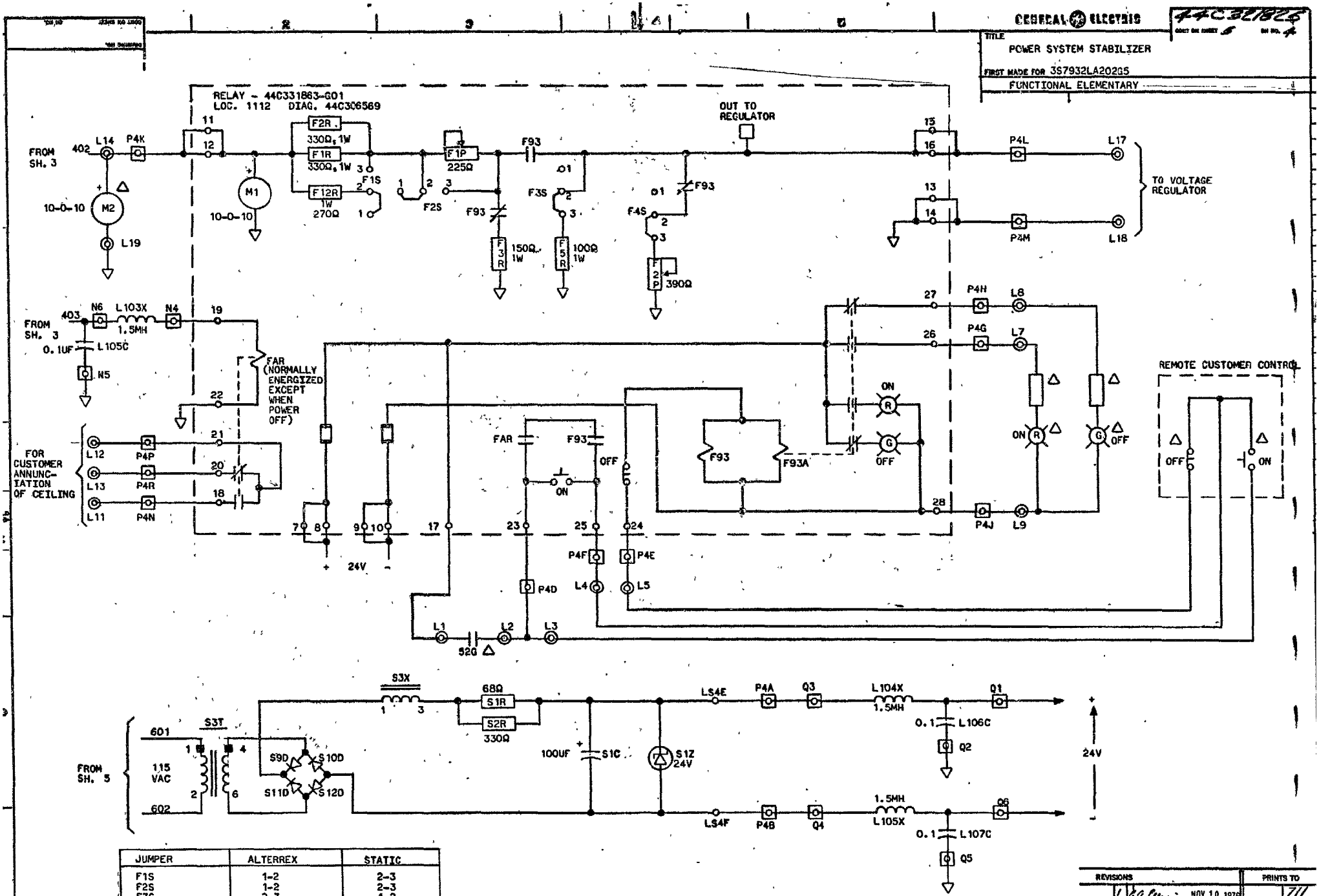


Figure 4, Sheet 3



JUMPER	ALTERREX	STATIC
F1S	1-2	2-3
F2S	1-2	2-3
F3S	2-3	1-2
F4S	2-3	2-3
POTENTIOMETER		
F1P	225	0
F2P	390	102

Figure 4, Sheet 4

REVISIONS		PRINTS TO
1	E.A. Clemens NOV 10 1976	777
2	H.T. Lyon MAY 9 1977	

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C321825

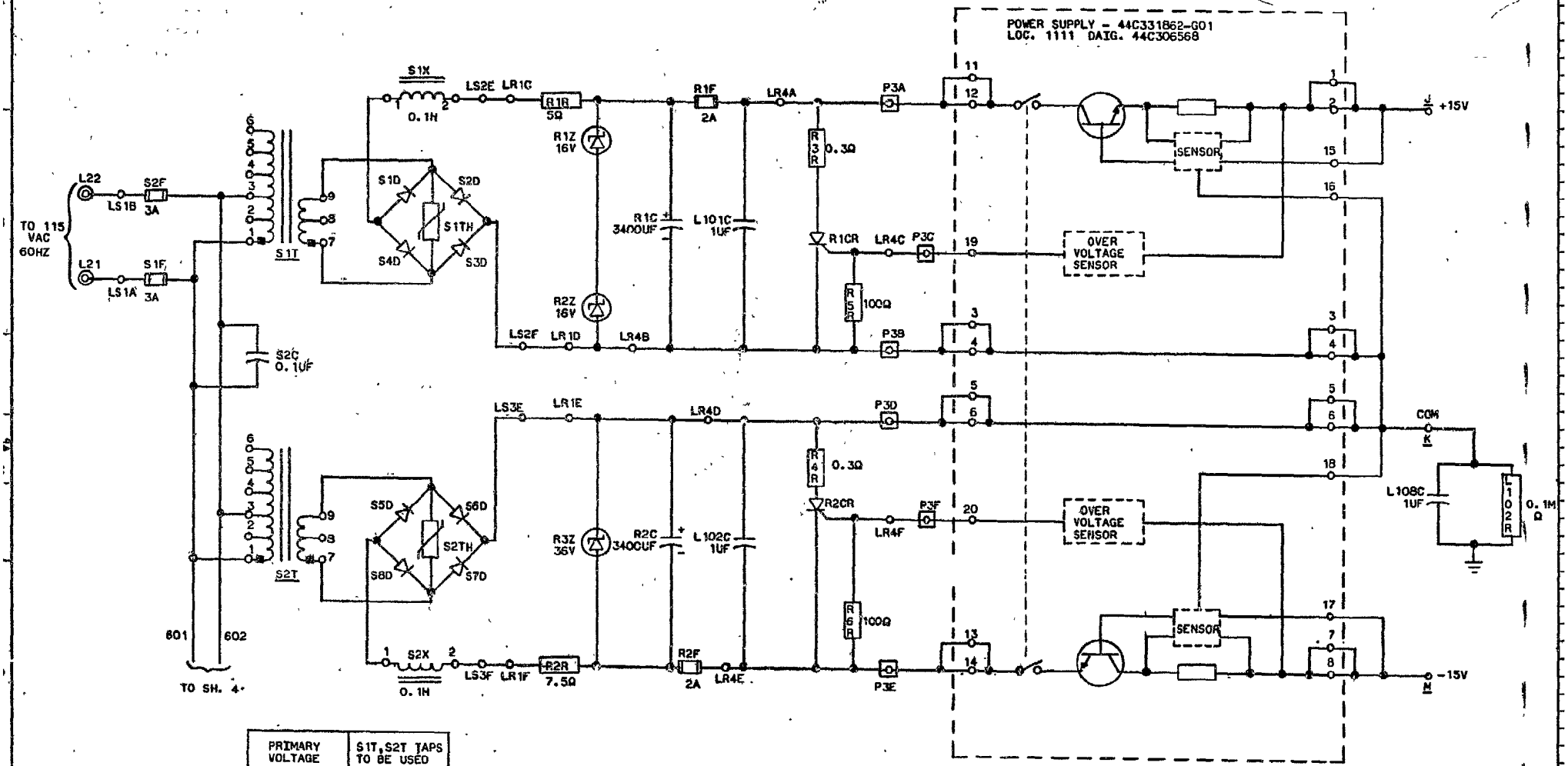
GENERAL ELECTRIC

44C321825 CONT ON SHEET 6 SH NO. 5

TITLE
POWER SYSTEM STABILIZER

FIRST MADE FOR 3S7932LA20265

FUNCTIONAL ELEMENTARY



PRIMARY VOLTAGE	S1T,S2T TAPS TO BE USED
110	1-2
115	1-3
120	1-4
125	1-5
135	1-6

Figure 4, Sheet 5

REVISIONS		PRINTS TO
1	P.A. Cerny NOV 10 1976	711

ISSUED OCT 2 2 1976

DRIVE SYSTEMS PROD DEPT WAYNESBORO, VA.

44C321825 CONT ON SHEET 6 SH NO. 5

TITLE
POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932LA202G5
FUNCTIONAL ELEMENTARY

CIRCUIT	SHEET
CEILING SENSING	3
POWER SUPPLY	
±15 VOLT	5
24 VOLT	4
RELAY	4
SIGNAL CONDITIONING	3
SPEED TRANSDUCER	2
WASHOUT AND OUTPUT	3

- NOT FURNISHED BY GE WAYNESBORO
- MOUNTED ON CUSTOMER CONTROL BOARD
- CUSTOMER TERMINAL BOARD
- INTERNAL TERMINAL BOARD POINT
- REDUNDANT PC BOARD PINS
- TERMINAL BOARD POINTS ON BACK OF PC BOARD RACK
- INDICATOR LIGHTS
- LIGHT EMITTING DIODE (LED)
- RELAY OR CONTACTOR COIL
- RELAY CONTACT
- RESISTOR
- POTENTIOMETER
- CAPACITOR
- INDUCTANCE, TRANSFORMER WINDING
- FUSE
- DISCONNECT SWITCH
- RHEOSTAT, ADJUSTABLE RESISTOR
- RECTIFIER (DIODE)
- THYRISTOR (SCR)
- ZENER DIODE
- OPERATIONAL AMPLIFIER (OP AMP)
- TRANSISTOR
- PUSHBUTTON (N.O.)
- PUSHBUTTON (N.C.)
- CIRCUIT COMMON
- TWISTED SHIELDED LEADS
- AND — AND CIRCUIT
- PCB MOUNTED SELECTOR SWITCH
- METAL OXIDE VARISTOR (M.O.V.)

REVISIONS

PRINTS TO 711

Figure 4, Sheet 6

GENERAL ELECTRIC 44C3085

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE: SPEED TRANSDUCER

FIRST MADE FOR 4800HZ ± 5%

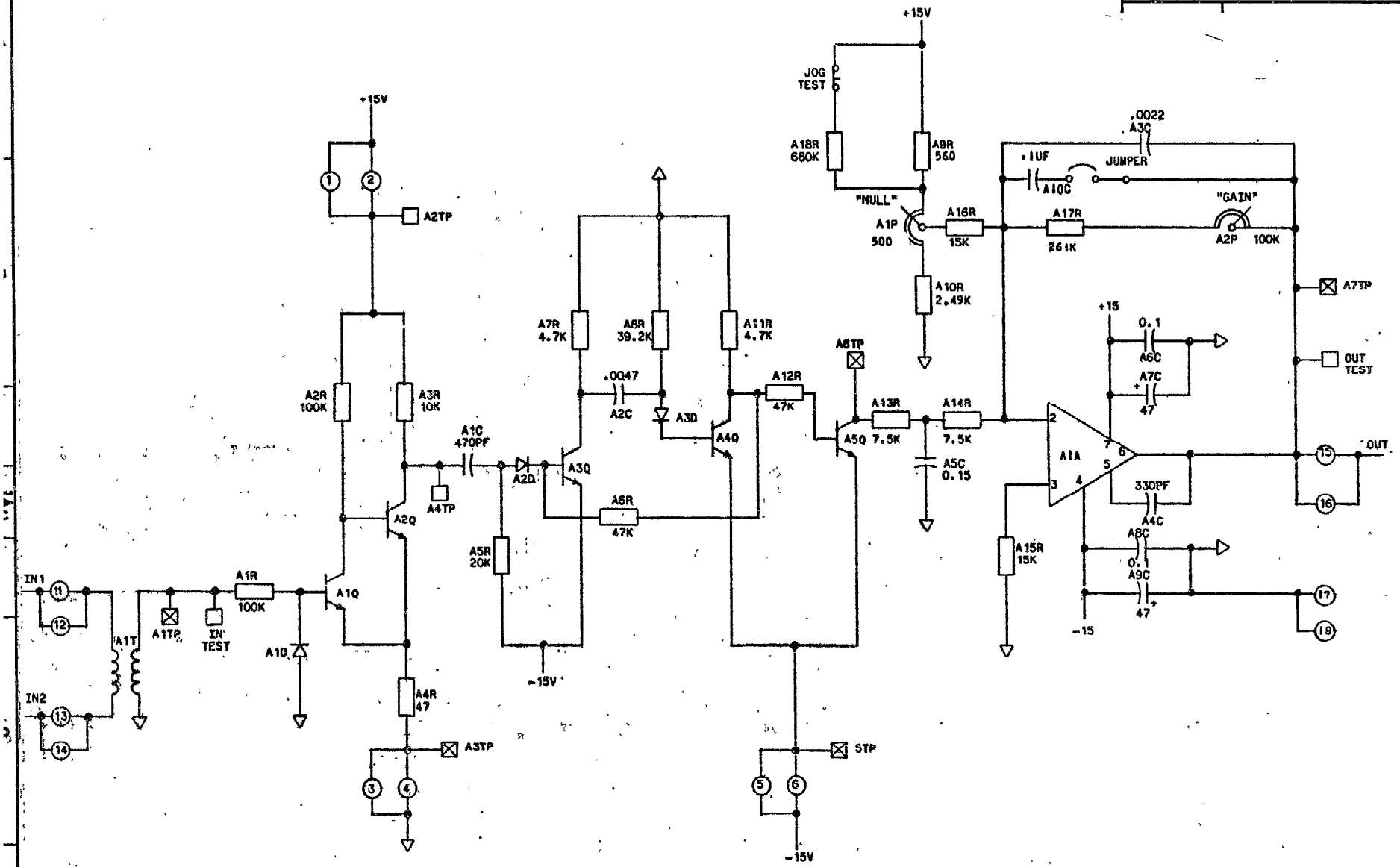


Figure 5

REVISIONS			PRINTS TO
1	E.A. Ellis 2-6-1976		11
2	E.A. Ellis 4/5/6 1976		1
3	DEC8 1975		

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C3085 88

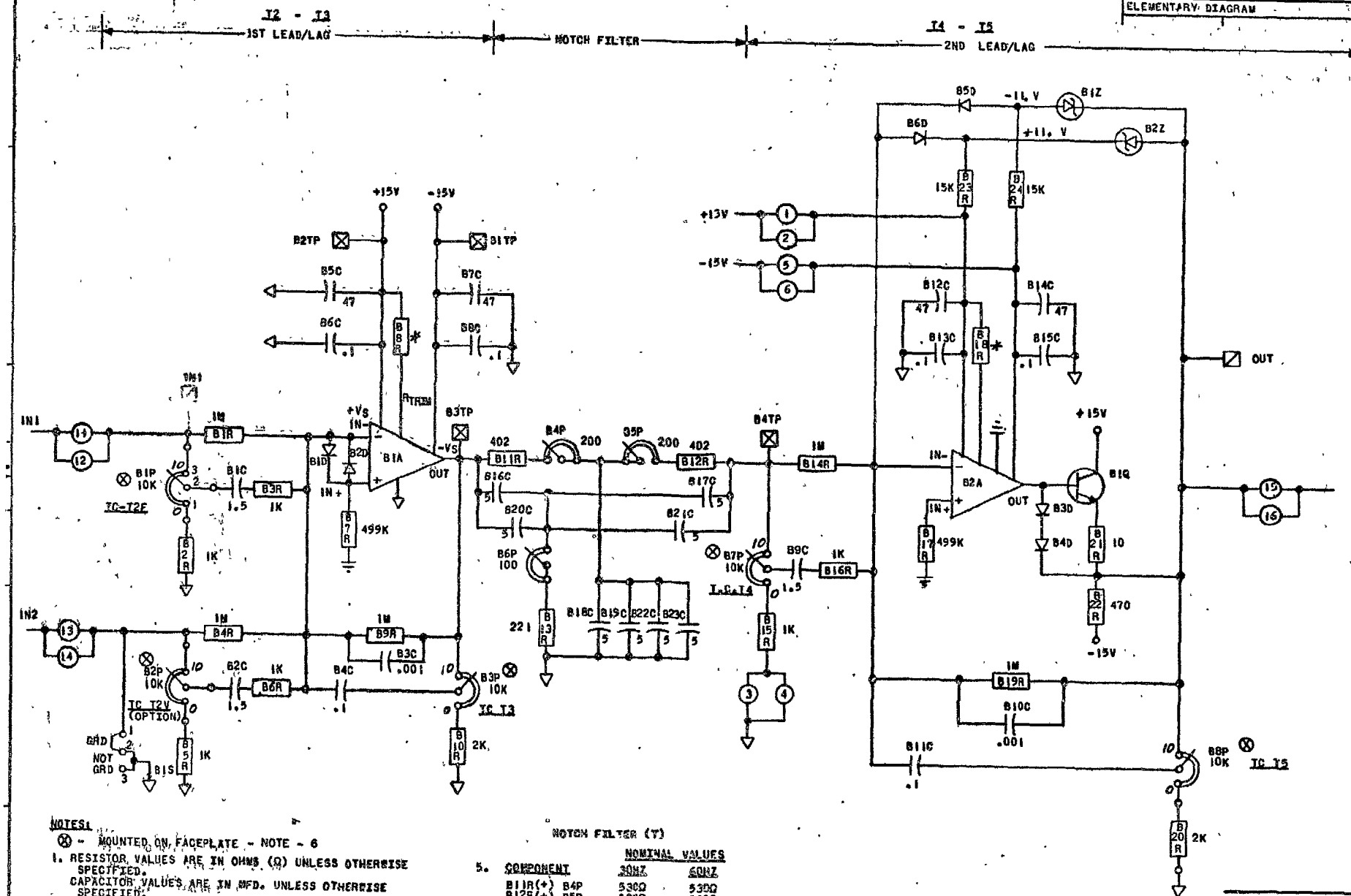
GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE: SIGNAL CONDITIONING POWER SYSTEM STABILIZER

FIRST MADE FOR 3S7932MA323G1, LA202Q1.

ELEMENTARY DIAGRAM



NOTES:

⊗ - MOUNTED ON FACEPLATE - NOTE - 6

1. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED. CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.
2. OMIT B20C, B21C, B22C, B23C ON 3600 RPM GENERATORS (60HZ).
3. *, RESISTOR VALUE WILL CHANGE FROM ONE UNIT TO NEXT.
4. ☒ BOARD MOUNTED TEST POINT.
6. ☒ TEST BNC CONNECTORS ON FACEPLATE OF BOARD.

NOTCH FILTER (T)

5.

COMPONENT	NOMINAL VALUES 30HZ	60HZ
B11R(+) B4P	530Ω	530Ω
B12R(+) B5P	530Ω	530Ω
B13R(+) B6P	265Ω	265Ω
B16C	10MFD	5MFD
B17C	10MFD	5MFD
B18C, B19C, B22C, B23C	20MFD	10MFD

Figure 5

REVISIONS		PRINTS TO
#	E.C.	JAN 21 1976

ISSUED	APPROVALS	DRIVE SYSTEMS PROD. DEPT.	44C306563
R. Vargas May 19, 1973	T. Smith	WAYNESBORO, VA.	

GENERAL ELECTRIC 44C306565

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
WASHOUT & OUTPUT
POWER SYSTEM STABILIZER
FIRST MADE FOR 357932MA325G1
ELEMENTARY DIAGRAM

WASHOUT | AMPLIFIER | STATIC SWITCH | OUTPUT STAGE

NOTES:

1. RESISTOR VALUES ARE IN OHMS (Ω), 1/2W UNLESS OTHERWISE SPECIFIED.
2. CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.
3. * - RESISTOR VALUE WILL VARRY WITH EACH UNIT.
4. JUMPER C4S 1-2, WHEN C3S IS IN 1-2 POSITION
5. JUMPER C4S 2-3, WHEN C3S IS IN 2-3 POSITION
6. ☐ TEST BNC CONNECTORS ON FACEPLATE OF BOARD.

Figure 5

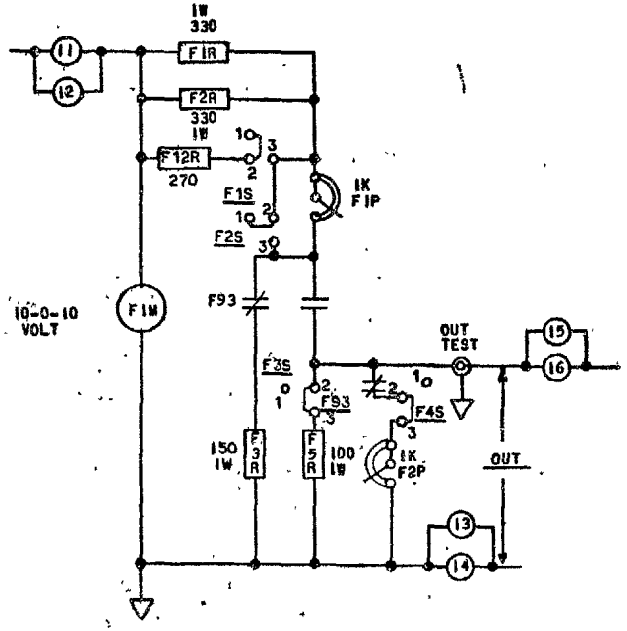
REVISIONS			PRINTS TO
1	T. Smith	Dec. 18, 1972	711
2	[illegible]	Feb 6, 1974	769 3
3	[illegible]	MAY 30 1975	
4	E.G. Ellis	JAN 21 1976	
5	E.G. Ellis	SEP 16 1976	
6	[illegible]	NOV 16 1976	

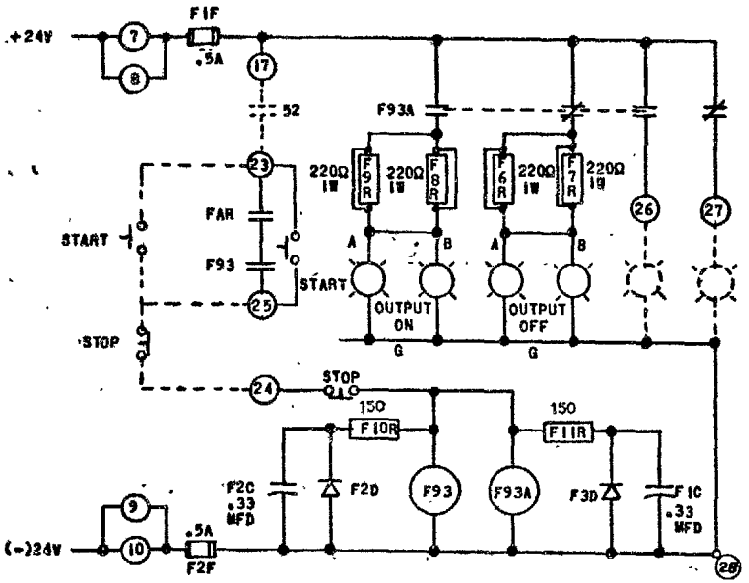
DRAWN BY E. Ellison 4/17/73 — APPROVED T. Smith
ISSUED R. Mayes May 18, 1973
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C306565 CONT ON SHEET SH NO

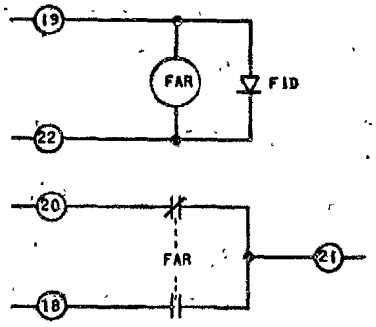
GENERAL ELECTRIC 44C306569

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES OR MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
RELAY
POWER SYSTEM STABILIZER
FIRST MADE FOR 3S7932LA202A1, MA329G1
ELEMENTARY DIAGRAM







JUMPERS USED TO CONNECT CORRECT IMPEDANCE TO VARIOUS REGULATORS.

JUMPER	GENERREX	ALTERREX	143	STATIC	ALTHYREX
F1S	1-2	1-2	1-2	2-3	1-2
F2S	2-3	1-2	1-2	2-3	1-2
F3S	1-2	2-3	1-2	1-2	1-2
F4S	2-3	2-3	2-3	2-3	2-3
F1P POTENTIOMETER	-0-	225Ω	158	-0-	305
F2P POTENTIOMETER	-0-	390	323	102	570

NOTES:

RESISTOR VALUES ARE IN Ω UNLESS OTHERWISE SPECIFIED.

Figure 5

REVISIONS		PRINTS TO
1	GA Elmore Aug 13, 1973	11
2	AN 33-8661 W Lunsford Dec 28, 1973	
3	FEB 17 1978	
4	GA Elmore NOV 11 1978	

MADE BY IRV MAYER 4/7/72 ISSUED Pr Troyer March 18, 1976
APPROVALS
DRIVE SYSTEMS PROD. DEPT. DIV. OR DEPT. WAYNESBORO, VA. LOCATION
44C306569 CONT OR SHEET SH NO

44C306569 CONTRACT NO CONT ON SHEET SH NO RFV NO

GENERAL ELECTRIC

44C306567

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
CEILING SENSING
FIRST MADE FOR MA327
ELEMENTARY DIAGRAM

COMPARATORS — LOGIC CIRCUITS — OUTPUT STAGE

+15V, -15V, IN, INPUT, OUT

E1TP, E2TP, E3TP, E4TP, E5TP, E6TP, E7TP, E8TP, E9TP, E10TP, E11TP

POS. LIMIT, NEG. LIMIT, POS. REF., NEG. REF.

E1A, E2A, E1L, E2L, E3L ONE SHOT

E1P 500K TIME DELAY SECONDS(S)

E1S TIME DELAY

IN RELAY SECTION

E1Q, E2Q, E3Q

E1IP RESET

E1P TIME DELAY

70 60 50 40 30 20 10 1 SECONDS

0 2 4 6 8 10 POT POSITION

E1S JUMPER 2-3

E1S JUMPER 1-2

NOTES:

1. ↷ - SHOWN INCREASE CLOCKWISE ROTATION
2. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED.
3. CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.
4. ☒ TEST BNC CONNECTORS ON FACE PLATE.

Figure 5

REVISIONS

1	E A Ellison	Sept 6 1973	
2	W Tea	OCT 9 1 1975	
3	E A Ellison	JAN 9 1 1976	
4	D. Swilley	JUL 21 1976	

PRINTS TO 711 769-3

MADE BY E A Ellison 4/29/73
ISSUED D Mayer May 21, 1973
APPROVED T. Smith
ERIE SYSTEMS PROD DEPT. OF WAYNESBORO, VA. LOCATION
44C306567 CONT ON SHEET SH NO

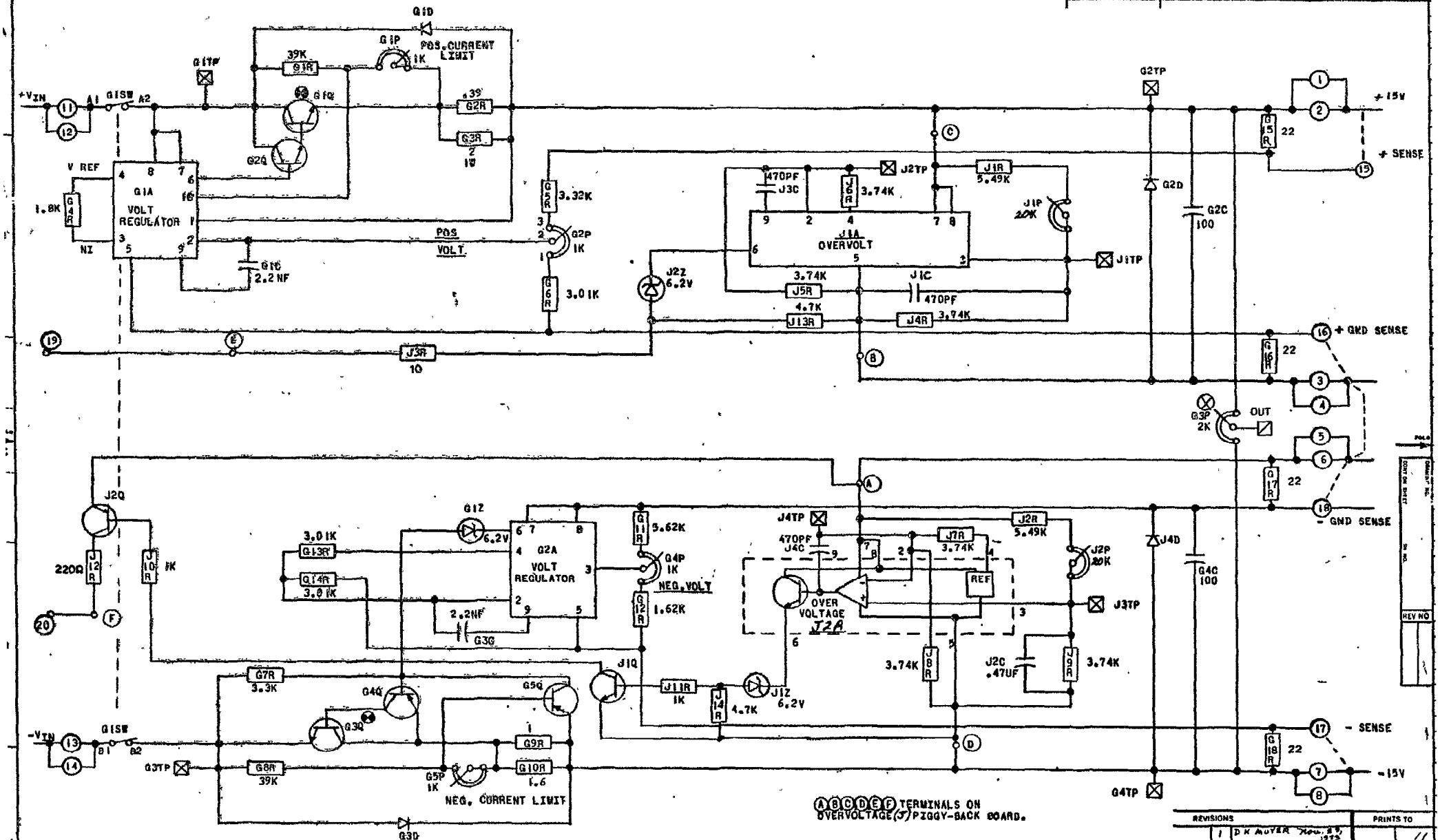
DRAWING NO 44C306567 CONT ON SHEET SH NO REV NO

GENERAL ELECTRIC

44C306568

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
POWER SUPPLY
FIRST MADE FOR 3S7932MA328Q1, LA202Q1
ELEMENTARY DIAGRAM



NOTES:
⊗ = MOUNTED ON FACEPLATE NOTE 3. ⊠ MOUNTED ON HEATSINK
1. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED. CAPACITOR VALUES ARE IN MFD'S - UNLESS OTHERWISE SPECIFIED.
2. BEFORE ENERGIZING PANEL, TURN J1P, J2P TRIM POT TO MAX OHM, POSITIVE AND NEGATIVE SUPPLIES. AFTER REGULATION IS ESTABLISHED, ADJUST G2P, G4P FOR 18 VOLTS ON BUS. DECREASE J1P, J2P
2. CONT'D. UNTIL FUSE BLOWS. RE-SET G2P, G4P FOR ± 15V. BEFORE RE-ENERGIZING, RE-SET G2P, G4P (10 TURNS CCW) TO LOWER REGULATED VOLTAGE. (FACTORY ADJUSTED)
3. ☐ TEST BNC CONNECTOR MOUNTED ON FACEPLATE.

Figure 5

REVISIONS			PRINTS TO
1	D K Auer Nov. 8, 1972		11
2	[illegible] Oct. 8, 1973		7643
3	P.A. Ellis JAN 21 1976		

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C306568

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS, DECIMALS, ANGLES

TITLE: UNDER- OVERVOLTAGE DETECTION

FIRST MADE FOR

ELEMENTARY DIAGRAM

REFERENCE AMPLIFIER | COMPARATOR | LOGIC CIRCUIT

+15V -15V

IN2 10K D5R D3D D4D D2C .22 100K D10R 3.9K D14R 10K D13R 10K D11R

IN1 +30V D1TP 806 D1R 402 D2R INPUT VOLTAGE D1P 100 D4R 10K D1C .5 D3R 200 GRD2

D2D D1D D1A TRIM RTRIM * D15R 4.99K D26R 10K D6R

5.6K D16R D8C .1 D9C .1 D5D D6D D7C 2.2 5.1K D17R D3P 2K NEG. LIMIT

D2A 220 D18R D4P 500 NEG. REF. D2Z 6.2V 820 D19R +15V D5TP +5V 5.1K D28R D8TP

D3C .1 D4C 47 D5C .1 D6C 47 D3TP D4TP

-15V 680 D7R 221 D8R 6.2V IN825 D1Z D2P 500 D2TP -5V 1.62K D9R REF.

OUT1 5.6K D20R D11C .1 D12C .1 D10C 22 5.1K D21R D5P 2K POS. LIMIT

D3A 220 D22R D6P 500 POS. REF. D3Z 6.2V 820 D23R +15V D6TP -5V D7TP 5.1K D29R

+15V 150Ω D24R D4Z 4.7V 10K D25R D13C 33 D9D D9TP

D10TP ONE SHOT D1L D11TP D2L D3L D12TP D13TP D14C 220pF

OUTPUT OUT2 D27R 220 OUT2 (TEST)

NOTES:
- ⊗ - MOUNTED ON FACEPLATE
- RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED.
- CAPACITOR VALUES ARE IN MFD. UNLESS OTHERWISE SPECIFIED.
- * - RESISTOR VALUE WILL VARY WITH EACH UNIT.
- ☒ BOARD MOUNTED TEST POINT.

Figure 5

REVISIONS: 1 gdi — Feb 6, 1974

PRINTS TO: 11, 769

MADE BY: [illegible] 4/18/73 ISSUED: D. Meyer May 21, 1973 APPROVALS: T. Smith

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA LOCATION

GAIN AND PHASE VS. FREQUENCY FOR PSS (EXCLUDING INPUT TRANSDUCER)

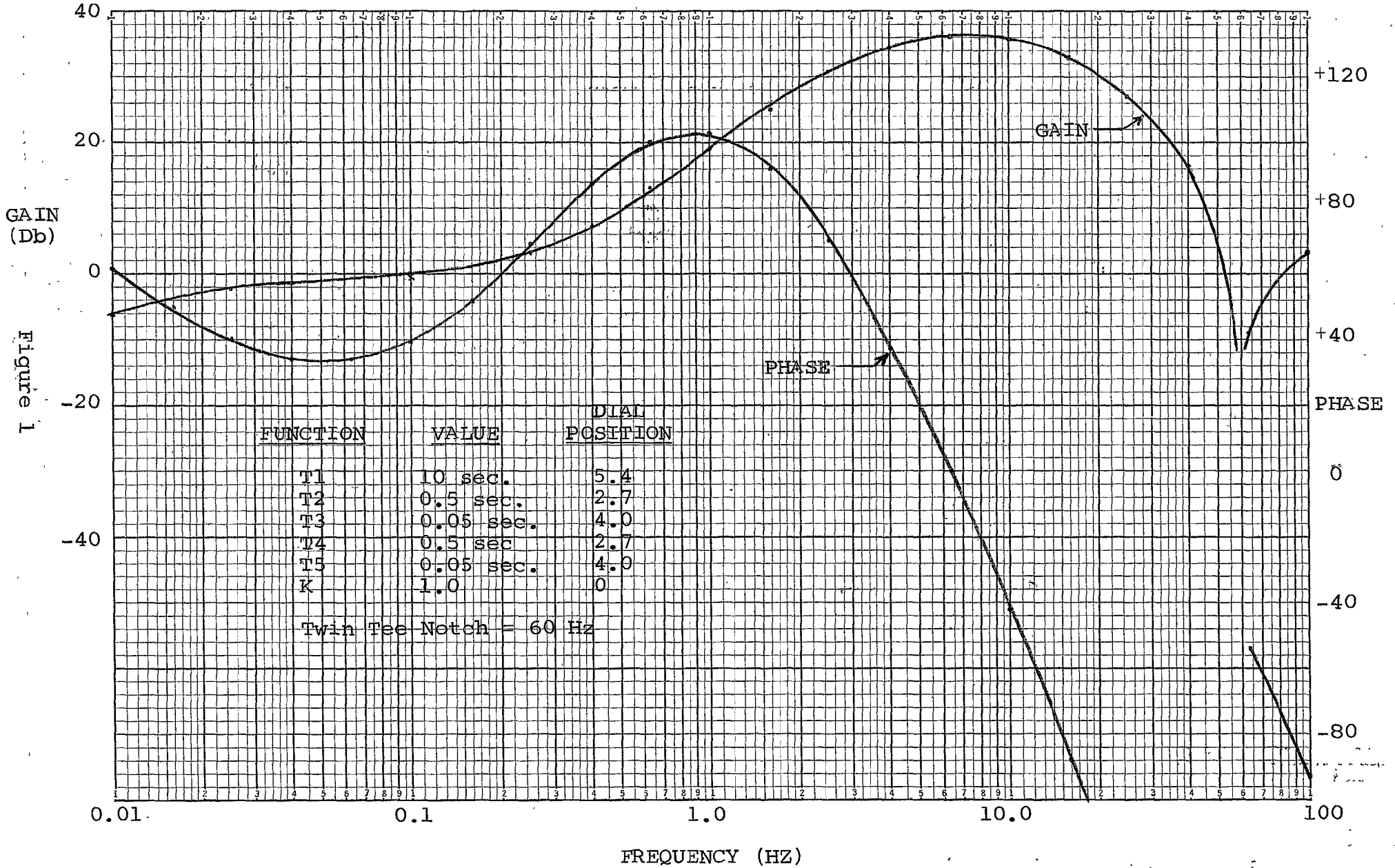


Figure 1